

Raghuram Kalyanam

Senior Data Scientist/Consultant



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Senior Data Scientist / AI Engineer working on production LLM systems : RAG architecture and retrieval quality, evaluation frameworks and LLM-as-judge pipelines, prompt architecture, and AI security and guardrails. Experience taking GenAI systems from prototype into monitored production in regulated and public-sector environments. Earlier background in time-series forecasting, reinforcement learning and MLOps across finance, automotive, aviation and construction.

Work Experience

October 2025
Present

Senior Data Scientist / Consultant, ELCA Informatique SA, Zurich

- > Developed a sovereign, fully self-hosted GenAI assistant platform for the **Canton of St. Gallen** handling documents classified confidential and secret, with purpose-specific assistants for writing, research and summarisation, using Open WebUI, Bifrost, vLLM and Gemma.
- > Designed prompt architecture for rule-following assistants, including compression of large official rule sets into stable system prompts.
- > Developed an internal multilingual staff assistant over a large corpus of work-instruction manuals for a **leading Swiss collection agency**, using Chainlit, OpenSearch, LiteLLM and Qwen3.
- > Designed and tuned lexical retrieval with per-language analysers, German compound-aware query shaping and LLM-generated section tags, and multimodal retrieval surfacing relevant screenshots without a vision model.
- > Built evaluation frameworks with DeepEval and MLflow, combining deterministic rule metrics, cross-family LLM judges and retrieval-versus-generation failure attribution.
- > Owned AI security and guardrails for a public-facing chatbot : layered defence against prompt injection and abuse, adversarial test corpora, and penetration-test remediation with regression coverage; mapped OWASP Top-10 for LLM Applications and data-protection requirements onto implemented controls.
- > Productionised the platform layer : zero-downtime reindexing, Keycloak OIDC SSO with directory-group RBAC, Prometheus and Grafana observability with per-use-case cost attribution, and GitLab CI/CD.
- > Evaluated cross-encoder reranking and vector search, and documented the decision to retain lexical retrieval for a curated German corpus.

LLMs RAG OpenSearch Open WebUI LiteLLM Bifrost Ollama vLLM Qwen3 Gemma DeepEval MLflow
Chainlit FastAPI Keycloak Prometheus Grafana Docker GitLab CI/CD Azure OpenAI Python
Prompt Engineering LLM-as-Judge Guardrails

July 2022
September 2025

Senior Data Scientist, Mercedes-Benz Mobility AG, Stuttgart

- > Built an LLM-powered semantic query system letting collections staff identify high-risk customers in natural language, using Model Context Protocols to translate intent into weighted SQL and vector queries across risk history, payment profiles and behavioural data.
- > Built a semantic fraud reasoning system with hybrid search over analyst comments and structured financial metrics; used FAISS to cluster historical fraud patterns and LangGraph to orchestrate multi-step, traceable risk assessments.
- > Integrated Whisper to transcribe and semantically index customer conversations, making them searchable by tone, intent and risk.

LLMs LangChain LangGraph MCPs ChromaDB pgvector FAISS Superlinked Whisper Ollama LiteLLM
OpenAI API Hugging Face FastAPI Docker PySpark SQL

January 2020 May 2022	Data Scientist / Consultant, Lufthansa Industry Solutions GmbH, Raunheim ➤ Developed an AI forecasting platform with rolling predictions, automated retraining and Bayesian hyperparameter optimisation, plus the ETL workflows enriching data with external regressors. ➤ Modelled resource and scheduling problems as reinforcement learning tasks solved with DQN variants; containerised training and inference on Docker and Kubernetes for production. ➤ Consulted aviation, automotive and public-sector clients; mentored junior data scientists and advised on ML architecture. PythonPyTorchTensorFlowscikit-learnProphetskforecastMLflowFastAPIDockerKubernetesAzure TerraformPySparkSQL
May 2015 October 2019	Research Associate, TU Kaiserslautern LivingLab SmartOfficeSpace, Kaiserslautern ➤ Developed reinforcement learning models controlling adaptive glass transparency to optimise visual and thermal comfort in real office environments, with a purpose-built daylight simulation environment generating synthetic training data and large-scale tuning via Ray Tune. ➤ Collaborated with SBRC (University of Wollongong, Australia) on a co-simulation framework for building performance and user comfort; six peer-reviewed publications. PythonPyTorchTensorFlowKerasRLlibRay TuneOpenAI GymFlaskSparkHadoopSQLite
November 2012 February 2015	Software Engineer, Ericsson R&D Center, Chennai, India ➤ Implemented VoLTE (Java) and Diameter (C++) protocols for telecom billing systems in an agile team; node performance improvements, on-call support for international customers, and release management. JavaC++Design PatternsPostgresWiresharkJenkinsJiraGit

</> Skills

Programming Languages	Python, SQL, Java, C++.
GenAI / LLM	LangChain, LangGraph, MCPs, OpenAI API, Azure OpenAI, LiteLLM, Bifrost, Ollama, vLLM, Open WebUI, Chainlit, Qwen3, Gemma, Mistral, LLaMA, Hugging Face Transformers.
RAG & Retrieval	RAG architecture, OpenSearch, BM25 & lexical analysers, hybrid search, query expansion & rewriting, reranking, chunking strategies, multimodal retrieval, ChromaDB, pgvector, FAISS, Superlinked, Docling, Scrapy, BeautifulSoup.
Evaluation & Quality	DeepEval, LLM-as-judge, GEval, golden datasets, deterministic rule metrics, MLflow experiment tracking, retrieval failure attribution, A/B experiments, prompt architecture.
AI Security	Prompt-injection guardrails, OWASP Top-10 for LLM Applications, adversarial test corpora, groundedness & hallucination controls, Keycloak OIDC, RBAC, DPIA-driven controls.
ML & Optimisation	PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, CatBoost, RLlib, Prophet, skforecast, Ray Tune, HyperOpt, Bayesian optimisation.
Cloud & MLOps	Azure, Docker, Docker Compose, Kubernetes, Helm, Terraform, GitLab CI/CD, Prometheus, Grafana, FastAPI, Flask, Git.
Data & Visualisation	PySpark, Pandas, NumPy, Power BI, Tableau, Matplotlib, Seaborn, Plotly, Streamlit, Whisper.

🎓 Education

2007 - 2012 **M.Sc Mathematics and Computing**, IIT Dhanbad, India.

🏆 Achievements

- Six peer-reviewed publications in reinforcement learning and building-performance modelling (see below).
- Ranked in the top 1% of 400,000 candidates in the IIT-JEE entrance examination.
- Held a merit scholarship throughout my studies; recognised as “Rockstar” at Ericsson for performance.

🗣️ Languages

English	Fluent
German	C1

Publications

- Kalyanam, Raghuram, and Sabine Hoffmann. 2021. "A Reinforcement Learning- Based Approach to Automate the Electrochromic Glass and to Enhance the Visual Comfort" *Applied Sciences* 11, no. 15 : 6949.
- Kalyanam, Raghuram, and Sabine Hoffmann. "Reinforcement learning based approach to automate the external shading system and enhance the occupant comfort." *AI in AEC conference*, Helsinki, Finland, March 2021.
- Kalyanam, Raghuram, and Sabine Hoffmann. "A Novel Approach to Enhance the Generalization Capability of the Hourly Solar Diffuse Horizontal Irradiance Models on Diverse Climates." *Energies* 13.18 (2020) : 4868.
- Kalyanam, Raghuram and Sabine Hoffmann. 2016. "Visual and thermal comfort with electrochromic glass using an adaptive control strategy." Paper presented at the *15th International Radiance Workshop (IRW)*, Padua, Italy, August 2016.
- Boudier, Katharina, Massimo Fiorentini, Sabine Hoffmann, Raghuram Kalyanam, and Georgios Kokogiannakis. 2016. "Coupling a thermal comfort model with building simulation for user comfort and energy efficiency." *In Proceedings of the Central European Symposium on Building Physics (CESBP) and BauSIM*, Dresden, Germany, September 2016, 481–487.
- Hoffmann, Sabine, Andrew McNeil, Eleanor S. Lee, and Raghuram Kalyanam. 2015. "Discomfort glare with complex fenestration systems and the impact on energy use when using daylighting control." *In Proceedings of the Advanced Building Skins Conference*, Bern, Switzerland, November 2015.